# C PROGRAMMING

*Lesson 1 — Introduction & Basic Syntax*

Based on: Bro Code C Tutorial Series

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## 1. Structure of a C Program

Every C program must follow this basic structure:

```
#include <stdio.h>

int main() {
    // your code goes here
    return 0;
}
```

### #include <stdio.h>

This is called a preprocessor directive. It loads the standard input/output library, which gives you access to printf(). Always place it at the very top of your file.

### int main()

This is the entry point of your program — the point where execution begins. Every C program must have a main() function. Without it, your program will not compile.

### return 0;

This tells the operating system that your program finished successfully. The value 0 means "no errors". Always include this at the end of main().

> **Remember:** Without the main() function, your program will not run and will throw an exit status 1 error.

## 3. Printing Output

Use printf() to display text on the screen. It comes from the stdio.h library.

```
printf("Hello, World!\n");
printf("This is line two.\n");
```

### Key Rules for printf()

- **Use \n** inside the string to move to a new line (like pressing Enter)
- **Every statement** must end with a semicolon ; — missing it causes a compiler error

- **The text** to print must be inside double quotes " "

## 4. Comments

Comments are notes in your code that the compiler completely ignores. They do not affect the program output. Use them to explain what your code does.

### Single-line comment

```
// This is a single-line comment
printf("Hello");  // This comment is after code
```

### Multi-line comment

```
/*
   This is a multi-line comment.
   Use it for longer explanations.
*/
```

> **Good Habit:** Add comments to explain what each section of your code does. It helps you when you look at old code later.

## 5. Key Rules to Always Remember

- **C is case-sensitive:** main is not the same as Main
- **Every statement** ends with a semicolon ;
- **Code inside main()** runs top to bottom, one line at a time
- **\n inside a string** = new line in the output

## 6. Practice Problem

**Practice: Your Info Card**

Write a C program that prints a simple info card about yourself.

Requirements:

- Print your name, age, and favourite hobby — each on a separate line
- Add a single-line comment above each printf() explaining what it prints
- Add a multi-line comment at the very top describing what the program does

### Expected Output (example)

```
Name: Alex
Age: 17
Hobby: Gaming
```